Microbes & cells — overview

Immune cells, pathogens and body cells side by side

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The microscope-style images are not real photographs — they are stylised drawings that hint at typical shape and staining. The infographic and kids styles are likewise hand-drawn SVGs.

Immune cells — Lymphocytes

The specialised troops of adaptive immunity.

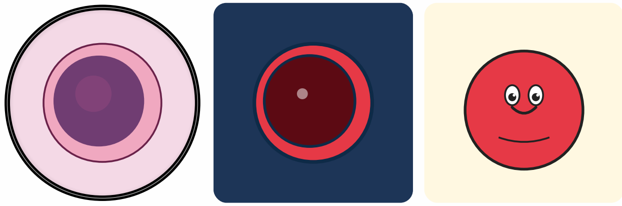
Helper T cell (CD4)

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Conductor of the immune response: recognises antigen fragments shown by other cells and releases messengers that switch other cells on.

DEPENDS ON / TARGETS: Needs antigen-presenting cells (dendritic cells, macrophages). Activates B cells and CD8 killer cells.

Cytotoxic T cell (CD8)



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Killer cell: spots infected or tumour cells by their MHC-I signal and kills them precisely with perforin and granzymes.

DEPENDS ON / TARGETS: Primed by dendritic cells, often with help from CD4 T cells. Targets virus-infected body cells and tumour cells.

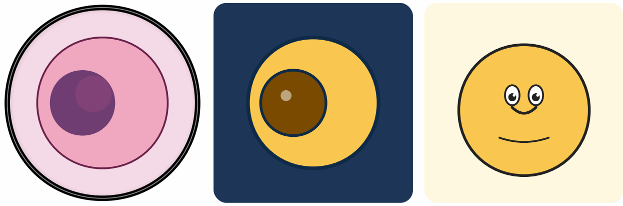
B cell

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Antibody factory: binds antigen with its B-cell receptor and matures into plasma cells that pump out tailored antibodies.

DEPENDS ON / TARGETS: Usually needs help from CD4 T cells to switch into high gear. Targets free bacteria, virus particles and toxins in blood and tissue.

Plasma cell

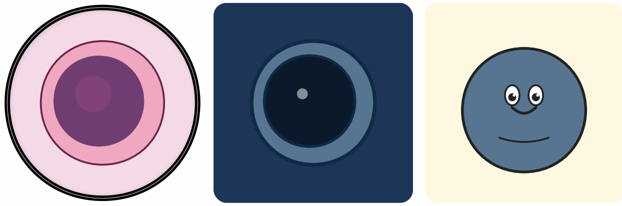


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: End stage of a B cell: a walking assembly line that secretes antibodies by the bucketload.

DEPENDS ON / TARGETS: Differentiates from an activated B cell. Its antibodies tag pathogens for other immune cells to deal with.

Natural killer cell (NK)

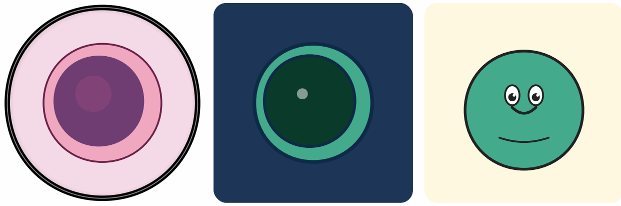


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Innate-immune bouncer: kills cells that have lost their MHC-I 'ID badge' — typical for virus infection or cancer.

DEPENDS ON / TARGETS: Revved up by interferons and IL-12. Kills virus-infected and tumour cells without prior priming.

Regulatory T cell (Treg)



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FUNCTION: Brake pedal of the immune system: damps other immune cells so the response doesn't overshoot and turn into autoimmunity.

DEPENDS ON / TARGETS: Acts on helper T cells, CD8 killers and B cells — slows them down rather than killing them.

Immune cells — Phagocytes & Granulocytes

The rapid-response squad of innate immunity.

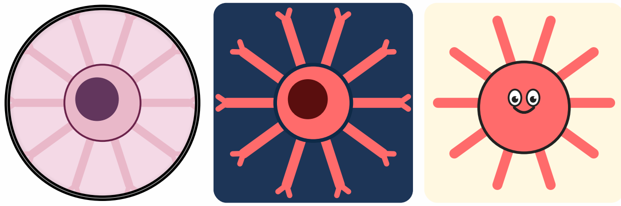
Macrophage

MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: 'Big eater': engulfs bacteria, dead cells and debris, presents fragments to T cells and calls reinforcements with cytokines.

DEPENDS ON / TARGETS: Differentiates from blood monocytes. Phagocytoses bacteria, fungi, cancer cells and cleans up wounds.

Dendritic cell



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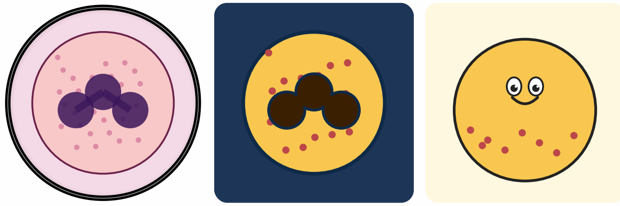
FUNCTION: Scout and storyteller: gathers antigen in tissues, travels to lymph nodes and shows it to T cells — kicks off the adaptive response.

DEPENDS ON / TARGETS: Bridges innate and adaptive immunity. Activates helper T cells and CD8 killer cells.

Neutrophil

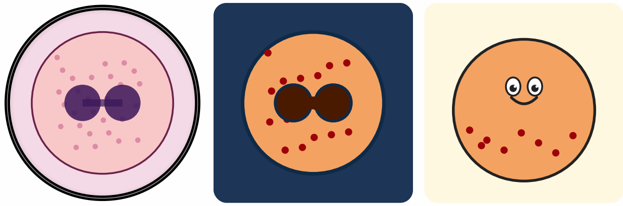


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: First responder: arrives first at sites of inflammation, eats bacteria, dumps antimicrobials and even casts sticky DNA nets (NETs).

DEPENDS ON / TARGETS: Chemokine-guided. Targets bacteria and fungi; lives only hours to a few days.

Eosinophil



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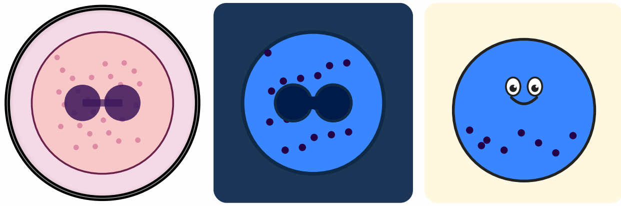
FUNCTION: Specialist against parasitic worms and a player in allergic reactions. Packs big red granules full of toxic proteins.

DEPENDS ON / TARGETS: Recruited by IL-5 from Th2 and ILC2 cells. Targets parasites and amplifies asthma and allergic reactions.

Basophil



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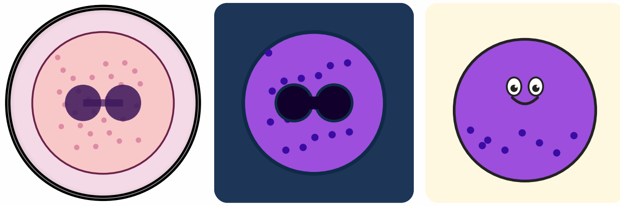
FUNCTION: Rare in blood but loud: releases histamine and heparin and drives allergic and inflammatory reactions.

DEPENDS ON / TARGETS: Responds to IgE antibodies. Involved in allergies and in defence against parasites.

Mast cell



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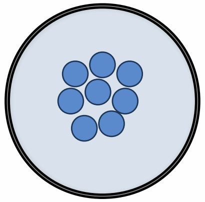
FUNCTION: Tissue-resident sentinel packed with granules: on alarm it dumps histamine in an instant — swelling, itching, help signals.

DEPENDS ON / TARGETS: Triggered by IgE antibodies and pathogen signals. Drives allergy and anaphylaxis reactions.

Pathogens

What the immune system has to deal with.

Cocci (round bacteria)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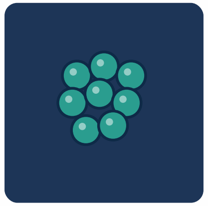
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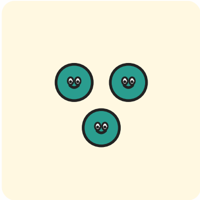
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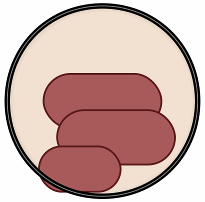
FUNCTION: Round bacteria, often in clusters or chains (Staphylococci, Streptococci). Some are harmless guests, others cause pneumonia, skin or throat infections.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and macrophages; antibodies help tag them.

Rod bacteria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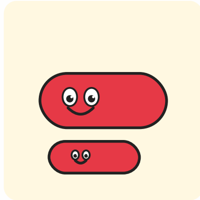
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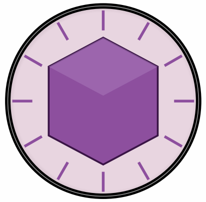
FUNCTION: Elongated bacteria like E. coli, Salmonella, tuberculosis bacilli. Multiply fast and can release toxins.

DEPENDS ON / TARGETS: Phagocytosed; intracellular species (TB) need T-cell help to be cleared.

Virus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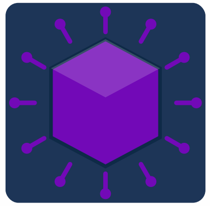
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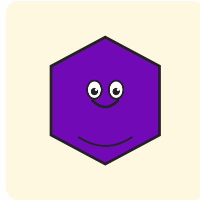
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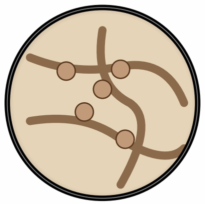
FUNCTION: Not really alive: just genetic material in a shell. Sneaks into body cells and forces them to build new virus particles.

DEPENDS ON / TARGETS: Hunted by CD8 T cells and NK cells; antibodies bind free virus particles.

Fungus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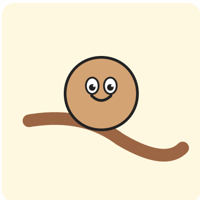
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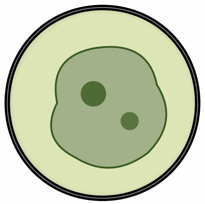
FUNCTION: Eukaryotic microbes — yeasts or filaments (hyphae). Examples: Candida in mouth/gut, Aspergillus in lungs.

DEPENDS ON / TARGETS: Controlled by neutrophils, macrophages and Th17 cells.

Parasite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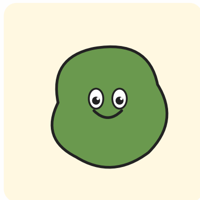
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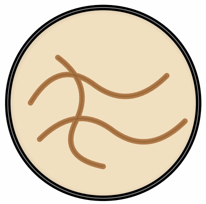
FUNCTION: From single-celled (Plasmodium → malaria) to worms. Lives in or on its host and damages it in the process.

DEPENDS ON / TARGETS: Often fought with IgE, eosinophils and mast cells; antibodies and Th2 responses are central.

Prion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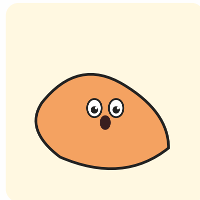
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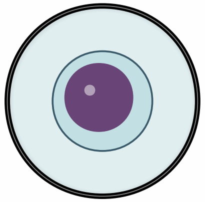
FUNCTION: Not alive, not a virus — a misfolded protein that forces other proteins to misfold the same way. Causes diseases like Creutzfeldt-Jakob, BSE.

DEPENDS ON / TARGETS: Largely invisible to the immune system; no antibody defence.

Hematopoietic lineage

All blood cells arise from a single stem cell in bone marrow.

Hematopoietic stem cell (HSC)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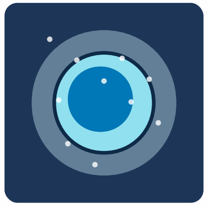
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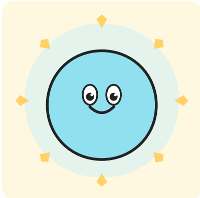
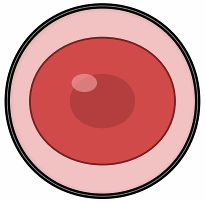

KIDS STYLE

FUNCTION: Mother of all blood cells. Lives in bone marrow, divides rarely, self-renews and gives rise to every blood lineage.

DEPENDS ON / TARGETS: Needs niche cells in bone marrow (osteoblasts, stromal cells) and growth factors.

Erythrocyte (red blood cell)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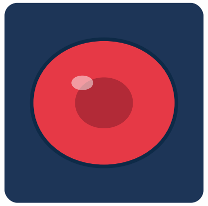
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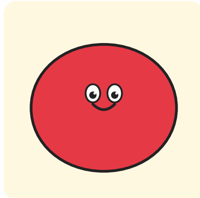
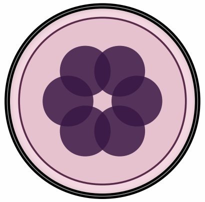

KIDS STYLE

FUNCTION: Oxygen carrier: no nucleus, packed with haemoglobin. Lives about 120 days and circulates millions of times.

DEPENDS ON / TARGETS: Production driven by the hormone erythropoietin from the kidney. Delivers oxygen to every body cell.

Megakaryocyte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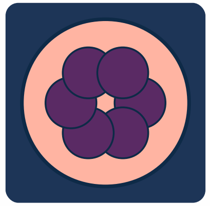
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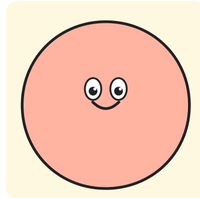
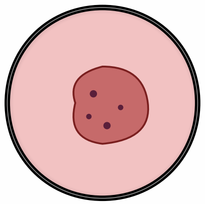

KIDS STYLE

FUNCTION: Giant bone-marrow cell with a multi-lobed nucleus. Pinches off thousands of platelets from its cytoplasm.

DEPENDS ON / TARGETS: Driven by thrombopoietin (TPO) from the liver. Provides the source material for platelets.

Thrombocyte (platelet)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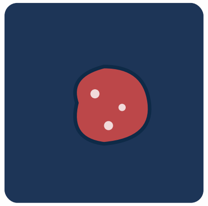
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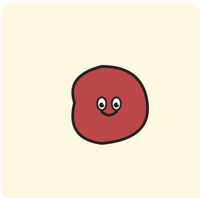
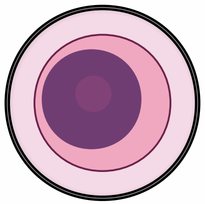

KIDS STYLE

FUNCTION: Tiny anucleate fragments. Plug wounds, form the first clot and trigger the coagulation cascade.

DEPENDS ON / TARGETS: Originate from megakaryocytes. Cooperate closely with coagulation factors and endothelial cells.

Monocyte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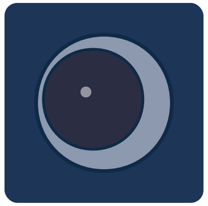
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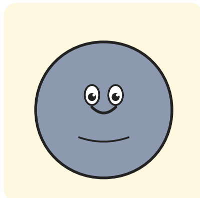
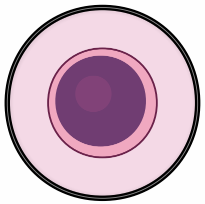

KIDS STYLE

FUNCTION: Large blood-borne precursor. Migrates into tissues and matures there into macrophages or dendritic cells.

DEPENDS ON / TARGETS: Responds to inflammatory signals. Supplies tissues with fresh phagocytes.

Lymphocyte (generic)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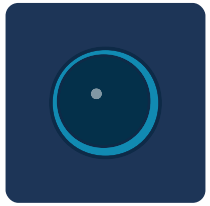
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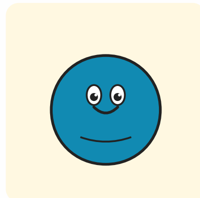

KIDS STYLE

FUNCTION: Umbrella term for T, B and NK cells. Small, round, with a dense nucleus — the accountants of the immune system.

DEPENDS ON / TARGETS: Mature in bone marrow (B, NK) or thymus (T). Patrol lymph nodes, spleen and blood.

Mesenchymal lineage

Connective tissues: bone, cartilage, muscle, fat.

Mesenchymal stem cell (MSC)

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Reserve cell for supporting tissues. Can turn into bone, cartilage, muscle, fat or connective-tissue cells.

DEPENDS ON / TARGETS: Lives in bone marrow, fat tissue and umbilical cord. Supplies precursors on demand.

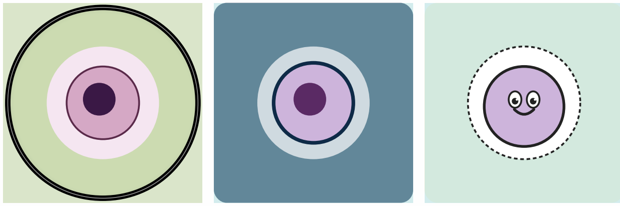
Osteoblast (bone builder)

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Builds new bone: secretes collagen and mineralises it with calcium. Later gets walled in as an osteocyte.

DEPENDS ON / TARGETS: Balances with osteoclasts that resorb bone. Hormonally controlled by PTH and vitamin D.

Chondrocyte (cartilage cell)

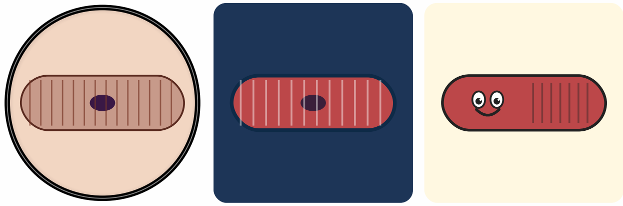


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Sits alone or in small groups inside a 'lacuna' of cartilage matrix. Produces and maintains the rubbery cartilage tissue.

DEPENDS ON / TARGETS: Cartilage has almost no blood supply — cells rely on diffusion. Crucial for joints, intervertebral discs, growth plates.

Myocyte (muscle cell)

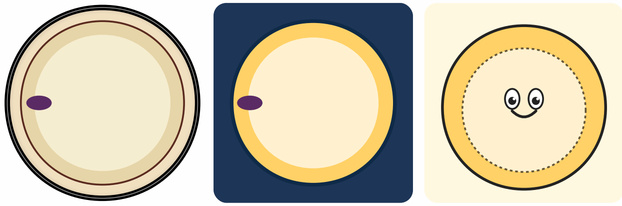


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Long, fibrous cell packed with actin/myosin filaments. Shortens on an electrical signal and generates force and movement.

DEPENDS ON / TARGETS: Commanded by motor neurons, fuelled by glucose and oxygen from the blood.

Adipocyte (fat cell)

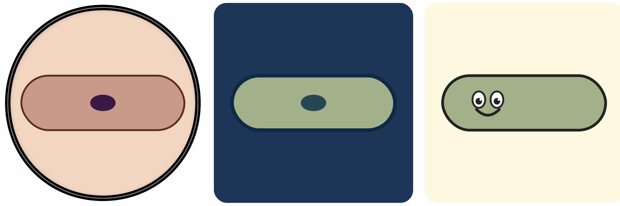


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Lipid storage: a single huge fat droplet pushes the nucleus and cytoplasm to the rim. Also acts as an endocrine cell (e.g. leptin).

DEPENDS ON / TARGETS: Stimulated by insulin to take up fat. Releases energy for other cells when food is scarce.

Fibroblast



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

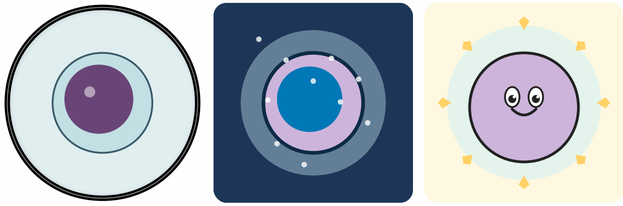
FUNCTION: Connective-tissue all-rounder: makes collagen and other fibres, keeps skin and organs in shape and closes wounds.

DEPENDS ON / TARGETS: Activated at wounds by growth factors (TGF-β, PDGF). Works closely with macrophages.

Neural lineage

Nervous system: processing, wiring and support.

Neural stem cell (NSC)



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Precursor of all neural cell types. In the adult brain only active in a few niches (e.g. hippocampus).

DEPENDS ON / TARGETS: Responds to growth factors (EGF, FGF) and to tissue cues.

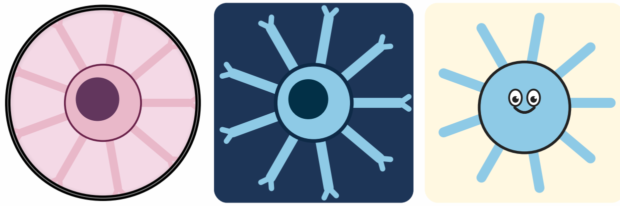
Neuron

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Compute-and-signal cell: collects inputs through dendrites, sums them up and fires electrical impulses down its axon.

DEPENDS ON / TARGETS: Depends on astrocytes (support) and oligodendrocytes (myelin). Communicates with other neurons through synapses.

Astrocyte

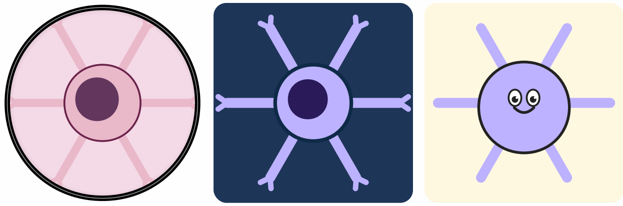


MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Star-shaped support cell. Feeds neurons, balances ions and neurotransmitters, helps form parts of the blood-brain barrier.

DEPENDS ON / TARGETS: Supports neurons and regulates local blood flow.

Oligodendrocyte



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Wraps several axons at once with myelin in brain and spinal cord — the electrical insulation of nerves.

DEPENDS ON / TARGETS: Partly fuels axons. In multiple sclerosis this myelin layer is the target.

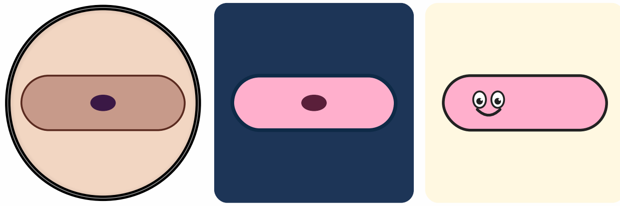
Microglia

MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

FUNCTION: Brain-resident macrophages: constantly probe the tissue with thin processes, clear debris, prune synapses.

DEPENDS ON / TARGETS: Yolk-sac derived in the embryo. First line of defence against brain infections.

Schwann cell



MICROSCOPE STYLE | INFOGRAPHIC STYLE | KIDS STYLE

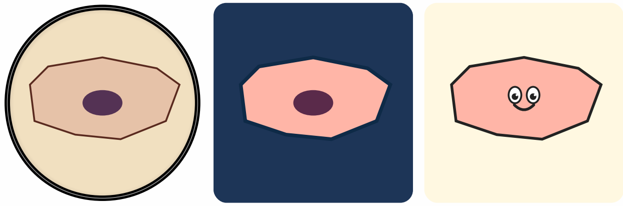
FUNCTION: Myelin maker in the peripheral nervous system: each Schwann cell wraps one segment of a single axon.

DEPENDS ON / TARGETS: Also helps peripheral nerves regenerate after injury.

Epithelial lineage

Body surfaces: skin, gut, lung.

Keratinocyte (skin cell)



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: Building block of the outer skin. Lays down keratin, cornifies upward and forms a waterproof, robust barrier.

DEPENDS ON / TARGETS: Renewed from stem cells in the basal layer. Cooperates with melanocytes and Langerhans cells.

Enterocyte (gut cell)

MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: Tall column-shaped cell with a dense brush border (microvilli). Absorbs nutrients from the gut and passes them on.

DEPENDS ON / TARGETS: Replaced every few days by intestinal stem cells. Forms a barrier against gut microbes.

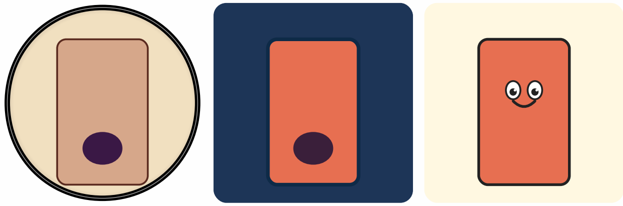
Goblet cell

MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: Goblet-shaped mucus producer in gut and airways. Releases mucin packets that form a protective gel over the epithelium.

DEPENDS ON / TARGETS: Cooperates with enterocytes. Protects against dryness and bacterial contact.

Paneth cell

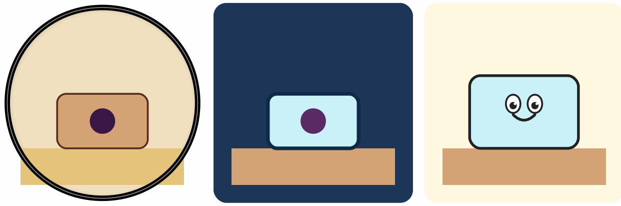


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: Sits at the base of intestinal crypts and fires antimicrobial proteins (defensins, lysozyme). Protects the intestinal stem cells.

DEPENDS ON / TARGETS: Together with stem cells forms the crypt-base niche.

Alveolar cell type II

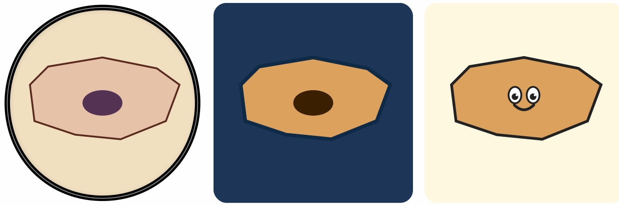


MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

FUNCTION: Small cuboidal lung cell. Makes surfactant — the film that keeps alveoli open. Also acts as stem cell for type I cells.

DEPENDS ON / TARGETS: Lives among the thin type-I cells that handle gas exchange.

Urothelial cell



MICROSCOPE STYLE INFOGRAPHIC STYLE KIDS STYLE

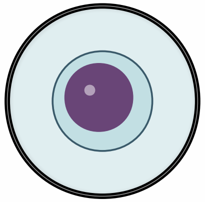
FUNCTION: Stretchable cell of the bladder wall. Reshapes its surface as the bladder fills and seals against acidic urine.

DEPENDS ON / TARGETS: Forms a barrier between urine and body. Renewed from basal cells.

Endothelial lineage

Inner lining of every blood and lymph vessel.

Endothelial progenitor (EPC)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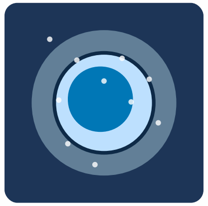
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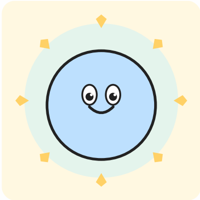
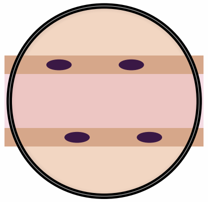

KIDS STYLE

FUNCTION: Young endothelial cell. Mobilised during wound healing and angiogenesis to replace damaged vessel wall.

DEPENDS ON / TARGETS: Recruited from bone marrow; responds to VEGF.

Endothelial cell (continuous)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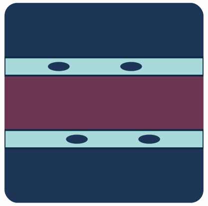
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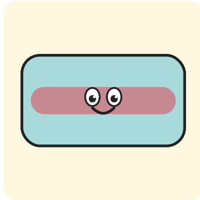
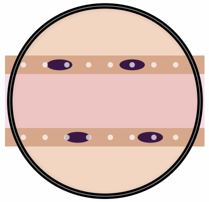

KIDS STYLE

FUNCTION: Flat cobblestone that lines most vessels. Regulates what diffuses between blood and tissue.

DEPENDS ON / TARGETS: Responds to blood-flow shear and hormones (e.g. histamine).

Fenestrated endothelium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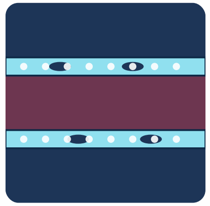
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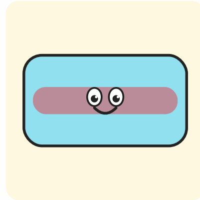
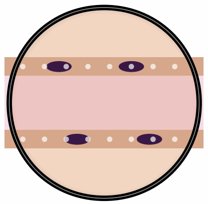

KIDS STYLE

FUNCTION: Endothelium with little 'windows' — permeable to fluid and molecules. Found in kidney, gut and endocrine glands.

DEPENDS ON / TARGETS: Filters blood, releases hormones quickly or takes up nutrients.

Sinusoidal endothelium (liver)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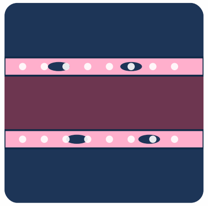
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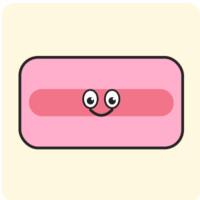
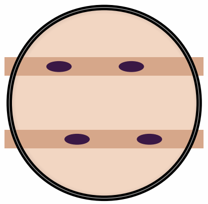

KIDS STYLE

FUNCTION: Very gappy — even large molecules pass through. Allows the liver direct contact with bloodflow.

DEPENDS ON / TARGETS: Works closely with hepatocytes and Kupffer cells (liver macrophages).

Lymphatic endothelium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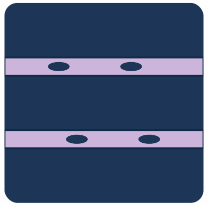
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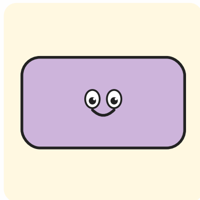
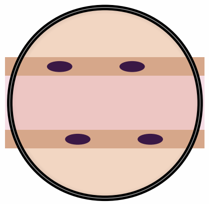

KIDS STYLE

FUNCTION: Wall of lymphatic vessels. The cells overlap like flaps so lymph flows only one way.

DEPENDS ON / TARGETS: Collects excess tissue fluid and returns it to the bloodstream.

Blood-brain-barrier endothelium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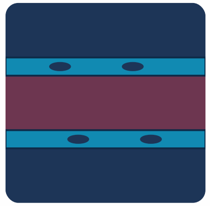
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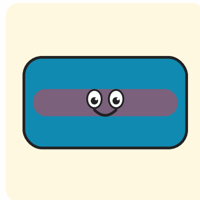

KIDS STYLE

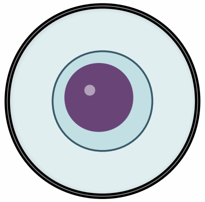
FUNCTION: Extra-tight: cells are locked together by tight junctions. Keeps almost everything out of the brain.

DEPENDS ON / TARGETS: Cooperates with astrocytes and pericytes that help form the barrier.

Induced pluripotent stem cells (iPS)

Lab-reprogrammed body cells — source for many therapies.

iPS cell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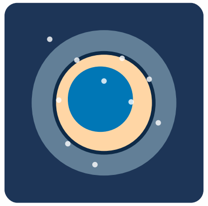
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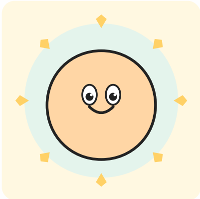
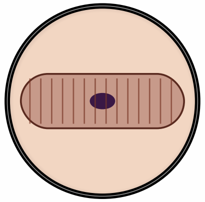

KIDS STYLE

FUNCTION: A body cell (e.g. skin) reset by a handful of transcription factors into an embryonic-like state — can develop into almost any cell type.

DEPENDS ON / TARGETS: Differentiated in vitro into a target lineage with specific growth factors.

Cardiomyocyte (heart muscle cell)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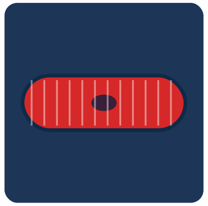
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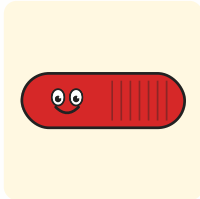
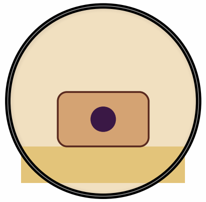

KIDS STYLE

FUNCTION: Striated muscle cell that beats rhythmically. Connected by intercalated discs so the heartbeat stays in sync.

DEPENDS ON / TARGETS: Responds to signals from the sinoatrial node and to hormones (adrenaline).

Hepatocyte (liver cell)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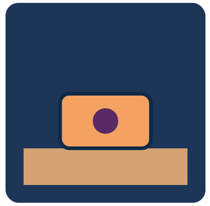
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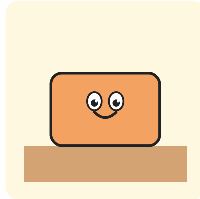
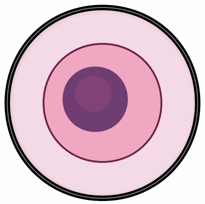

KIDS STYLE

FUNCTION: Metabolic all-rounder: breaks down toxins, makes bile and proteins, stores glucose as glycogen.

DEPENDS ON / TARGETS: Receives blood from portal vein (gut) and hepatic artery. Can divide and regenerate after damage.

Beta cell (pancreas)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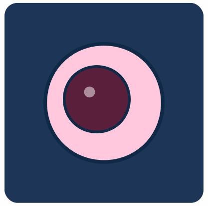
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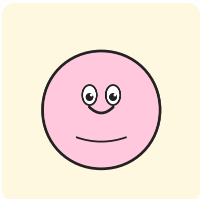
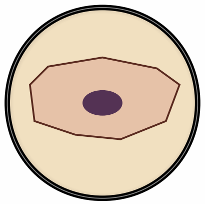

KIDS STYLE

FUNCTION: Insulin maker in the islets of Langerhans. Senses blood glucose and releases insulin when it rises.

DEPENDS ON / TARGETS: Destroyed by the immune system in type-1 diabetes — replacing it from iPS cells is an active research area.

Retinal pigment epithelium (RPE)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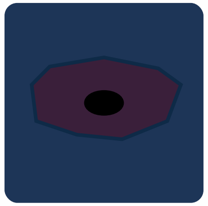
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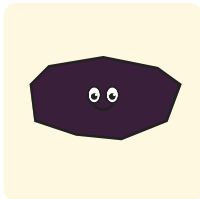
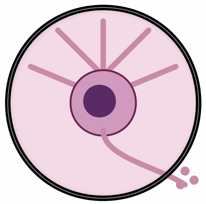

KIDS STYLE

FUNCTION: Dark cell layer behind the photoreceptors. Recycles spent receptor tips and supplies the photoreceptors with nutrients.

DEPENDS ON / TARGETS: Fails in age-related macular degeneration — iPS-derived RPE cells are being tested as transplants.

Dopaminergic neuron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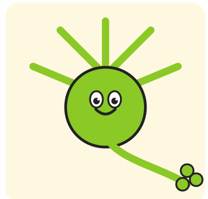
KIDS STYLE

FUNCTION: Neuron that releases dopamine. Crucial for reward, motivation and movement planning.

DEPENDS ON / TARGETS: Dies off in Parkinson's disease — iPS-derived dopamine neurons are being explored as cell-replacement therapy.